

Time series for the development of renewable energy sources in Germany

based on statistical data from the
Working Group on Renewable Energy-Statistics (AGEE-Stat)
(Status: February 2026)

For our environment



Umwelt 
Bundesamt

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The Working Group on Renewable Energy Statistics (AGEE-Stat) was established in February 2004 in order to place statistics and data relating to renewable energy sources on a comprehensive, up-to-date and coordinated basis (more information on: www.erneuerbare-energien.de/EE/ee-in-zahlen-arbeitsgruppe).

With the increasing share of renewable energy sources in the energy system and increasing national and international reporting obligations the demand on reliable data concerning the development of all renewable energy sources in Germany has been growing. Therefore, the coordination office of the Working Group engages in various research projects to improve the data basis and the scientific calculation methods- supported by the working group members (ministries, federal offices and academic institutions).

To analyze the historic development of the renewable energy sources in Germany since 1990 several data sources were evaluated and supplemented by (model-based) calculations. Wherever possible official data were used. Partially missing values were extrapolated from existing data. Single data gaps were closed by estimations, which is why the data are governed by a certain degree of uncertainties.

In some cases data are not available before a specific point of time and previous years could not be estimated reliable, but if new information allow conclusions to be drawn on historic values these data will be complemented or corrected. In addition, changes of the data collection methods or categories may result in breaks in time series, but only in cases in which a retrospective amendment is not possible and reasonable. Rounding may also cause discrepancies in the totals. Therefore the published data are partially preliminary and can differ in comparison to previous issues.

Detailed background information on data sources and methodology of the time series of the gross electricity production and installed electrical capacity can be found on:
www.umweltbundesamt.de/publikationen/datenquellen-methodik-der-agee-stat-zeitreihen-zur (available in German only).

Beside these time tables the AGEE-Stat publishes monthly and quarterly reports.

These can be found on:

www.umweltbundesamt.de/themen/klima-energie/erneuerbare-energien/erneuerbare-energien-in-zahlen/monats-quartalsberichte-der-agee-stat

Further enquiries to the data and methodology can be adressed on

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Table 1: Development of renewable energy sources

values before 2005 hidden

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Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Gross final energy consumption ¹⁾	188.019	227.802	261.788	267.720	264.252	305.858	316.109	351.782	364.416	358.666	387.587	389.345	420.834	434.532	456.282	472.063	472.075	497.437	502.015	522.670	540.508
Gross electricity production	63.538	72.795	89.999	95.106	96.974	106.446	125.648	145.091	153.677	163.740	190.073	191.107	217.674	225.291	243.590	253.543	239.684	255.600	275.641	288.084	290.150
Production of district heating	6.892	7.899	8.261	9.744	10.575	11.565	12.640	14.885	16.035	16.428	17.989	18.528	18.791	21.347	22.130	22.346	24.264	24.095	24.305	25.383	25.541
Final energy consumption for heating and cooling (including district heat) ²⁾	102.588	115.307	126.861	137.235	135.823	164.899	156.331	170.760	177.734	160.691	165.566	165.999	170.514	174.893	179.073	176.928	195.161	204.110	186.757	197.320	209.848
Final energy consumption for heating and cooling (without electricity and district heat) ³⁾	96.315	108.136	119.322	128.343	126.187	154.302	144.954	157.445	163.444	146.319	149.752	149.800	154.075	156.481	159.419	157.107	173.644	182.891	165.421	175.036	187.405
thereof in: Households	57.329	67.465	71.239	82.248	75.208	90.988	84.620	97.371	101.640	85.857	87.620	85.585	88.680	91.498	93.120	90.001	102.007	112.318	94.760	101.637	112.044
thereof in: TCS	14.441	16.806	19.516	24.092	26.432	31.861	26.591	29.197	31.174	28.864	31.670	31.952	33.323	33.552	34.983	36.029	38.813	37.090	38.515	39.669	41.631
thereof in: Industry	24.545	23.865	28.567	22.003	24.547	31.453	33.743	30.877	30.630	31.598	30.462	32.263	32.072	31.431	31.316	31.077	32.823	33.483	32.145	33.730	33.730
Final energy consumption in the transport sector ⁴⁾	22.627	40.456	45.968	36.226	32.441	35.622	35.361	37.223	34.277	35.348	33.296	33.642	34.623	35.994	36.041	44.304	39.850	41.375	44.751	43.389	47.741

Figures in [PJ]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Primary energy consumption ⁵⁾	788	962	1.116	1.157	1.154	1.317	1.369	1.534	1.584	1.576	1.682	1.687	1.798	1.832	1.909	1.974	1.970	2.050	2.054	2.127	2.182

differences in the totals may occur due to roundings

1) according to the German government's Energy Concept

2) sectoral local consumption of energy for application purposes of heating and cooling, without electricity, including district heat

3) sectoral local consumption of energy for application purposes of heating and cooling, without secondary energy carriers electricity and district heat

4) consumption of renewable-based fuels and electricity in the transport sector (excluding consumption in agriculture, forestry and military)

5) till 2002 by Working Group on Energy Balances (AGEB), calculated according to the "physical energy content" principle, since 2003 by AGEE-Stat based on JAQ-REN

Sources: AGEE-Stat based on AGEb [1]; Eurostat [18]; IEA [22]; EP/ER [16], [17] and further sources see table 3, 5 und 6; partially preliminary data

Table 2: Shares of renewable energy sources

values before 2005 hidden

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Figures in [%]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
of gross national electricity consumption	10,3	11,7	14,4	15,3	16,6	17,2	20,6	23,8	25,3	27,5	31,6	31,8	36,2	37,9	42,2	45,4	41,7	46,3	52,9	54,4	55,1
of district heat production ¹⁾	5,1	5,8	6,3	7,3	8,1	8,1	9,7	11,1	11,8	13,5	14,2	14,2	14,4	16,4	17,4	18,9	17,6	20,1	21,1	21,2	20,8
of final energy consumption for heating and cooling (including district heat) ²⁾	8,0	8,6	10,6	10,4	11,2	12,2	12,6	13,6	13,6	13,7	13,6	13,4	13,7	14,5	14,6	14,7	15,4	17,5	17,4	18,2	19,0
of final energy consumption for heating and cooling (without secondary energy carriers electricity and district heat) ³⁾	8,4	8,9	11,0	10,8	11,6	12,7	12,9	13,8	13,8	13,7	13,5	13,3	13,7	14,3	14,3	14,3	15,2	17,2	17,0	17,9	18,8
thereof in: Households	10,7	12,2	15,7	15,2	14,9	16,5	18,1	19,1	18,9	19,1	18,6	17,6	18,5	18,9	18,2	17,8	19,9	23,1	21,5	22,8	24,2
thereof in: TCS	7,5	7,3	10,8	11,6	13,5	15,2	13,9	15,5	14,7	15,0	15,9	16,8	17,3	19,8	20,3	20,9	21,9	22,5	25,4	26,3	26,0
thereof in: Industry	5,8	5,5	6,4	5,0	6,3	6,9	7,3	7,0	7,0	7,4	7,0	7,2	7,1	7,1	7,3	7,5	7,2	8,1	8,4	8,8	9,0
of final energy consumption in the transport sector	3,6	6,2	7,4	5,9	5,4	5,8	5,7	6,1	5,5	5,6	5,2	5,2	5,2	5,6	5,5	7,5	6,7	6,9	7,6	7,4	8,0

Figures in [%]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
of primary energy consumption	5,4	6,5	7,9	8,0	8,5	9,3	10,1	11,3	11,4	11,9	12,6	12,5	13,3	13,9	14,9	16,6	15,8	17,5	19,3	20,1	20,8

1) Net heat production, including grid losses

2) sectoral local consumption of energy for application purposes of heating and cooling, without electricity, including district heat

3) sectoral local consumption of energy for application purposes of heating and cooling, without secondary energy carriers electricity and district heat

Sources: see table 1 and 7; partially preliminary data

Table 2.1: Development of renewable energies according to international reporting conventions (SHARES Indicators)

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RED	Period of Validity [in Years]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Reporting based on	RED I																RED II				RED III
RES	Gross final energy consumption - Total	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Gross final consumption of energy from renewable sources [TWh]	188,8	228,7	256,7	268,1	271,0	311,7	316,6	350,7	364,7	362,1	382,3	388,6	401,4	432,5	446,4	465,4	485,7	510,2	498,7	514,9	555,4
	Total gross final consumption of energy [TWh]	2.635,0	2.701,1	2.557,2	2.661,4	2.497,5	2.671,4	2.538,9	2.588,3	2.650,8	2.517,5	2.565,0	2.609,7	2.630,5	2.595,8	2.585,6	2.437,8	2.516,2	2.449,5	2.318,2	2.290,9	2.332,7
	Share of renewables in gross final consumption of energy [%]	7,2	8,5	10,0	10,1	10,9	11,7	12,5	13,5	13,8	14,4	14,9	14,9	15,5	16,7	17,3	19,1	19,3	20,8	21,5	22,5	23,8
RES-E	Gross final energy consumption - Electricity	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Renewable electricity generation (normalized) [GWh]	64,7	73,6	84,8	91,1	101,4	111,6	126,3	142,0	151,9	165,7	181,5	191,8	206,0	220,1	230,8	243,1	250,1	262,2	271,0	286,1	301,3
	Total gross electricity consumption [GWh]	611,8	616,4	617,9	615,1	578,6	611,8	603,5	601,7	600,7	588,1	594,1	594,3	595,3	585,6	568,3	549,9	568,9	545,7	521,8	529,0	517,2
	Share of renewables in gross final consumption of electricity [%]	10,6	11,9	13,7	15,1	17,5	18,2	20,9	21,6	25,3	28,2	30,9	32,3	34,6	37,6	40,6	44,2	44,0	48,0	51,9	54,1	58,7
RES-HEC	Gross final energy consumption - Heating and cooling	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Gross final consumption of energy from renewable sources in the heating and cooling sector [TWh]	99,7	112,0	121,8	133,0	130,6	157,3	148,9	161,1	171,0	152,9	156,8	154,0	157,4	165,6	167,6	163,6	204,1	213,8	191,6	195,9	214,4
	Total gross final consumption of energy in the heating and cooling sector [TWh]	1.332,2	1.378,9	1.243,5	1.354,8	1.341,6	1.374,3	1.246,4	1.293,9	1.343,9	1.216,0	1.248,3	1.271,5	1.271,5	1.260,3	1.256,2	1.246,3	1.313,9	1.229,6	1.133,2	1.108,4	1.135,3
	Share of renewables in gross final consumption of energy in the heating and cooling sector [%]	7,7	8,4	10,2	10,3	11,2	12,1	12,6	13,4	13,4	13,4	13,4	13,0	13,4	14,2	14,5	14,5	15,5	17,4	16,9	17,7	18,9
RES-DHHC	Gross final energy consumption - District heating and cooling	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Gross final consumption of energy from renewable sources and from waste heat in district heating and cooling [TWh]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	24,2	24,0	24,3	25,3	25,3
	Total gross final consumption of energy in district heating and cooling [TWh]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	130,1	118,7	114,1	114,3	112,4
	Share of renewables in gross final consumption of energy in district heating and cooling [%]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	18,6	20,3	21,3	22,2	20,7
RES-T	Final energy consumption in the transport sector	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Final consumption of energy from renewable sources in the transport sector [TWh]	24,4	41,9	45,9	37,9	34,6	38,3	39,0	43,9	44,7	42,9	41,4	45,2	46,1	50,3	48,9	59,1	47,0	59,9	72,4	74,1	81,5
	Total final consumption of energy in the transport sector [TWh]	616,4	625,0	609,4	599,7	588,9	596,9	604,4	600,0	612,8	621,4	630,6	644,4	656,4	633,9	640,8	590,3	580,3	594,5	591,0	599,4	759,2
	Share of renewables in final consumption of energy in the transport sector [%]	4,0	6,7	7,5	6,3	5,9	6,4	6,5	7,3	7,3	6,9	6,6	7,0	7,0	7,9	7,6	10,0	8,1	10,1	12,3	12,4	11,3

Table 3: Gross electricity production from renewable energy sources

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Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Hydropower ¹⁾	19.638	20.031	21.170	20.443	19.031	20.953	17.671	21.755	22.998	19.587	18.977	20.546	20.150	18.098	20.135	18.721	19.657	17.625	19.895	22.481	16.897
Wind energy onshore	27.774	31.324	40.507	41.385	39.382	38.371	49.280	50.948	51.819	57.026	72.340	67.650	88.018	90.484	101.150	104.796	92.075	99.149	117.877	113.749	107.225
Wind energy offshore	0	0	0	0	38	176	577	732	918	1.471	8.284	12.274	17.675	19.467	24.744	27.306	24.373	25.124	23.887	26.064	26.458
Solar Photovoltaic	1.308	2.265	3.137	4.508	6.715	11.963	19.991	26.744	30.621	35.448	38.076	37.556	38.761	44.330	45.212	49.500	50.486	61.287	64.384	75.878	91.591
Solid biofuels ²⁾	7.478	8.819	8.699	9.296	9.746	10.351	10.517	10.694	10.555	10.797	11.034	10.798	10.644	11.068	11.038	11.306	10.738	10.663	10.460	10.259	10.126
Liquid biofuels	121	749	987	1.133	1.700	1.331	398	355	299	347	443	508	454	397	344	320	210	97	110	101	82
Biogas	1.803	3.557	8.915	11.648	14.020	16.265	19.936	25.920	27.468	28.614	30.086	30.726	31.089	30.266	30.106	30.671	31.011	30.970	28.272	29.268	27.785
Biomethane	0	0	21	46	81	385	596	1.117	1.713	2.512	3.147	3.146	2.966	2.980	2.990	3.046	3.273	3.098	3.120	2.796	2.713
Sewage gas	1.096	1.057	1.033	1.094	1.131	1.203	1.280	1.314	1.308	1.336	1.389	1.440	1.460	1.555	1.581	1.579	1.576	1.547	1.529	1.531	1.533
Landfill gas	1.068	1.092	1.009	864	788	674	628	536	483	435	396	358	338	305	287	247	249	206	211	198	158
Biogenic fraction of waste ³⁾	3.252	3.901	4.521	4.671	4.323	4.746	4.755	4.951	5.415	6.069	5.768	5.930	5.956	6.163	5.806	5.820	5.792	5.628	5.701	5.544	5.357
Geothermal energy	0,2	0,4	0,4	18	19	28	19	25	80	98	133	175	163	178	197	231	244	206	195	215	225
Total	63.538	72.795	89.999	95.106	96.974	106.446	125.648	145.091	153.677	163.740	190.073	191.107	217.674	225.291	243.590	253.543	239.684	255.600	275.641	288.084	290.150
for information: amount of electricity under the EEG ⁴⁾	43.967	51.545	67.010	71.148	75.377	82.332	103.136	118.330	125.693	136.936	162.730	162.348	188.300	196.306	212.765	221.956	203.362	219.765	235.565	238.062	-

1) river and storage power plants including pumped storage plants with natural inflow

2) including sewage sludge

3) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste

4) fixed remuneration, market premium (since 2012), "green electricity privilege" and other direct marketing as well as the amount of remunerated self consumed electricity of photovoltaic plants under the EEG regulations between 2009-2012, including mine gas

annual statement of account for the EEG of the transmission system operators: www.netztransparenz.de

Sources: AGEE-Stat based on StBA [26], [27]; BNetzA [8]; ÜNB; ZSW; DENA [24]; BDEW; VDEW [20]; AGEB [1]; DBFZ [13]; IE [21]; partially preliminary data

Table 3.1: Net electricity production from renewable energy sources

values before 2005 hidden

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Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Hydropower ¹⁾	19.344	19.718	20.811	20.134	18.743	20.682	17.326	21.332	22.660	19.310	18.665	20.215	19.985	17.926	19.957	18.546	19.483	17.458	19.722	22.280	16.724
Wind energy onshore	27.229	30.710	39.713	40.574	38.610	37.619	48.314	49.949	50.803	55.908	70.922	66.324	86.293	88.710	99.166	102.741	90.270	97.205	115.565	111.519	105.122
Wind energy offshore	0	0	0	0	38	174	568	722	905	1.449	8.162	12.092	17.414	19.179	24.379	26.903	24.013	24.752	23.534	25.679	26.067
Solar Photovoltaic	1.282	2.220	3.075	4.420	6.583	11.729	19.599	26.220	30.020	34.753	37.330	36.820	38.001	43.461	44.326	48.529	49.496	60.085	63.121	74.390	89.795
Solid biofuels ²⁾	6.721	7.913	7.824	8.285	8.733	9.247	9.395	9.509	9.392	9.657	9.855	9.673	9.542	9.807	9.790	10.002	9.594	9.525	9.385	9.179	9.063
Liquid biofuels	117	726	958	1.099	1.649	1.291	386	319	286	333	425	481	428	381	330	308	201	91	104	96	77
Biogas	1.749	3.450	8.647	11.298	13.599	15.777	19.338	25.143	26.644	27.756	29.184	29.804	30.156	29.358	29.206	29.754	30.084	30.044	27.427	28.393	26.954
Biomethane	0	0	20	44	79	376	582	1.091	1.672	2.452	3.071	3.070	2.895	2.930	2.923	2.977	3.195	3.024	3.045	2.729	2.648
Sewage gas	1.057	1.023	999	1.033	1.042	1.093	1.179	1.217	1.215	1.283	1.308	1.331	1.357	1.551	1.578	1.576	1.572	1.544	1.526	1.528	1.530
Landfill gas	1.040	1.046	962	826	713	600	595	512	461	416	384	343	324	279	262	233	234	193	198	184	147
Biogenic fraction of waste ³⁾	2.422	2.965	3.519	3.667	3.352	3.755	3.795	3.971	4.305	4.838	4.565	4.746	4.802	4.932	4.626	4.638	4.575	4.436	4.521	4.352	4.228
Geothermal energy	0	0	0	12	13	20	19	25	69	67	91	164	157	126	144	173	174	152	139	148	155
Total	60.961	69.771	86.528	91.392	93.154	102.363	121.096	140.010	148.432	158.222	183.962	185.063	211.354	218.640	236.687	246.380	232.891	248.509	268.287	280.477	282.510

1) river and storage power plants including pumped storage plants with natural inflow
2) including sewage sludge
3) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste

Sources: AGEE-Stat based on StBA [26], [27]; BNetzA [8]; ÜNB; ZSW; DENA [24]; BDEW; VDEW [20]; AGEB [1]; DBFZ [13]; IE [21]; partially preliminary data

Table 4: Installed electrical capacity of renewable energy plants

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Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Hydropower ¹⁾	5.210	5.193	5.137	5.164	5.340	5.407	5.625	5.607	5.590	5.580	5.589	5.629	5.627	5.347	5.396	5.454	5.489	5.569	5.607	5.496	5.497
Wind energy onshore	18.248	20.474	22.116	22.794	25.697	26.823	28.524	30.711	32.969	37.620	41.297	45.283	50.174	52.328	53.187	54.256	55.887	57.988	60.971	63.535	68.145
Wind energy offshore ²⁾	0	0	0	0	35	80	188	268	508	994	3.283	4.152	5.406	6.393	7.555	7.874	7.874	8.216	8.473	9.215	9.733
Solar Photovoltaic	2.055	2.901	4.169	6.119	10.567	18.007	25.914	34.077	36.710	37.900	39.224	40.680	42.291	45.209	48.863	54.471	60.203	67.971	84.082	102.364	119.953
Solid biofuels	1.218	1.411	1.431	1.457	1.470	1.502	1.554	1.558	1.623	1.589	1.592	1.600	1.601	1.567	1.568	1.571	1.587	1.592	1.590	1.615	1.658
Liquid biofuels	60	177	295	341	412	410	345	277	263	232	232	231	230	230	231	231	201	196	192	187	178
Biogas ³⁾	665	1.000	1.226	1.419	2.520	3.015	3.837	4.212	4.317	4.380	4.601	4.780	5.173	5.610	5.951	6.343	6.453	6.516	6.575	6.612	6.671
Biomethane ³⁾	0	0	6	16	18	96	218	256	383	603	614	653	567	602	603	621	629	630	641	690	698
Sewage gas ⁴⁾	161	170	177	186	192	200	233	236	240	245	245	245	242	421	396	374	373	371	368	369	369
Landfill gas	248	252	257	268	261	237	232	214	210	211	183	172	165	169	162	156	137	132	131	129	128
Geothermal energy	0	0	3	3	4	4	4	15	25	28	28	34	34	34	39	39	46	51	50	50	65
Biogenic fraction of waste ⁵⁾	587	637	614	684	720	762	743	714	930	944	962	978	1.004	1.063	1.084	1.024	1.040	1.023	1.004	996	996
Total	28.452	32.215	35.431	38.451	47.236	56.543	67.417	78.145	83.768	90.326	97.850	104.437	112.514	118.973	125.035	132.414	139.919	150.255	169.684	191.258	214.091

1) river and storage power plants including pumped storage plants with natural inflow

2) installed capacity of offshore wind energy plants connected to the network

3) since 2013 including additional capacity for increased flexibility of electricity production

4) till 2014 calculated based on the electricity production and full-load hours specific to the energy source, since 2015 net additions according to registry data of the Federal Network Agency (BNetzA), since 2018 based on StBA

5) the stated values represent an amount of 50% of the total installed capacity of thermal waste combustion plants using renewable and fossil waste. For the wohle time series 50% of the total capacity is considered as renewable.

Sources: AGEE-Stat based on BNetzA [8]; StBA [27]; ZSW, DENA [24]; BDEW; VDEW [20], DBFZ [13], DEWI [14]; IE [21]; partially preliminary data

Table 4.1: Additional differentiation Solar Photovoltaik

values before 2005 hidden

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gross electrical capacity	Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	building-based PV systems	1.906	2.688	3.783	5.513	9.167	14.991	20.491	25.248	26.801	27.532	28.228	29.097	30.197	32.205	35.009	38.932	42.465	47.167	58.944	70.775	80.023
	thereof: ≤ 2 kW ¹⁾	35	38	41	43	45	47	50	54	55	55	56	57	59	60	62	73	111	365	1.270	2.568	3.818
	thereof: $2 < x \leq 10$ kW	608	831	1.090	1.464	1.988	2.671	3.479	4.291	4.784	5.055	5.293	5.563	5.924	6.332	6.914	8.131	9.217	10.802	14.179	16.342	17.696
	thereof: $10 \leq x < 20$ kW	407	571	790	1.130	1.750	2.620	3.498	4.153	4.344	4.386	4.421	4.453	4.489	4.540	4.594	4.784	5.476	6.494	10.052	13.434	15.902
	thereof: $20 \leq x < 30$ kW	375	540	785	1.156	1.930	3.020	3.823	4.444	4.563	4.589	4.627	4.675	4.756	4.871	5.030	5.359	5.723	6.119	7.107	8.269	9.273
	thereof: $30 \leq x < 100$ kW	351	500	747	1.177	2.193	3.863	5.243	6.298	6.557	6.670	6.763	6.904	7.123	7.480	7.921	8.521	8.981	9.419	10.307	11.417	12.294
	thereof: $100 \leq x < 500$ kW	99	152	246	403	922	2.081	3.290	4.463	4.811	5.012	5.264	5.368	5.599	6.140	6.966	7.771	8.292	8.762	9.795	11.083	12.104
	thereof: $500 \leq x \leq 750$ kW	6	14	22	37	112	276	460	635	690	715	723	759	925	1.453	2.201	2.930	3.259	3.475	3.820	4.215	4.502
	thereof: $700 < x \leq 1.000$ kW	4	9	12	29	106	215	350	477	500	521	537	640	640	640	637	653	663	703	956	1.264	1.500
	thereof: > 1.000 kW	21	33	50	74	121	198	298	433	497	529	544	678	682	689	684	710	747	1.028	1.498	2.183	2.934
gross electrical capacity	Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	ground-mounted PV systems	149	213	386	606	1.400	3.016	5.423	8.829	9.909	10.368	10.996	11.583	12.094	13.004	13.854	15.539	17.734	20.804	25.138	31.589	39.930
	thereof: < 10 kW	7	9	10	10	10	13	13	14	14	14	14	14	14	15	15	16	16	18	22	24	25
	thereof: $10 \leq x < 30$ kW	2	3	4	5	6	8	8	9	10	10	10	10	10	11	11	12	12	13	17	22	26
	thereof: $30 \leq x < 100$ kW	20	22	23	25	30	34	38	43	45	47	48	48	49	51	53	56	60	63	70	80	87
	thereof: $100 \leq x \leq 750$ kW	31	44	60	87	150	272	372	539	601	632	659	669	771	1.116	1.468	1.892	2.137	2.257	2.363	2.503	2.640
	thereof: $750 < x \leq 1.000$ kW	7	10	14	29	69	128	193	335	392	415	439	451	456	460	464	488	540	573	698	970	1.370
	thereof: $1.000 < x \leq 10.000$ kW	82	125	275	440	1.028	2.242	4.105	6.222	7.207	7.636	8.216	8.782	9.187	9.725	10.215	11.135	12.375	14.232	16.010	18.311	20.937
	thereof: > 10.000 kW	-	-	-	10	107	319	694	1.667	1.640	1.614	1.610	1.609	1.607	1.626	1.628	1.940	2.594	3.648	5.958	9.679	14.845
	Total	2.055	2.901	4.169	6.119	10.567	18.007	25.914	34.077	36.710	37.900	39.224	40.680	42.291	45.209	48.863	54.471	60.203	67.971	84.082	102.364	119.953

net electrical capacity	Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	building-based PV systems	1.695	2.397	3.387	4.959	8.324	13.716	18.840	23.297	24.745	25.417	26.052	26.839	27.840	29.646	32.175	35.709	38.881	43.055	53.417	63.663	71.401
	thereof: ≤ 2 kW ¹⁾	33	35	38	40	42	44	46	50	51	51	53	54	55	56	58	68	101	310	1.009	1.930	2.651
	thereof: $2 < x \leq 10$ kW	545	746	979	1.316	1.795	2.426	3.181	3.943	4.404	4.654	4.873	5.120	5.451	5.826	6.364	7.490	8.491	9.957	13.101	15.115	16.380
	thereof: $10 \leq x < 20$ kW	357	502	697	1.002	1.566	2.364	3.188	3.800	3.980	4.020	4.052	4.081	4.115	4.162	4.212	4.385	4.994	5.874	8.933	11.853	14.003
	thereof: $20 \leq x < 30$ kW	331	479	701	1.040	1.757	2.770	3.521	4.104	4.216	4.240	4.275	4.320	4.395	4.501	4.648	4.949	5.279	5.633	6.516	7.560	8.436
	thereof: $30 \leq x < 100$ kW	310	445	669	1.062	1.999	3.546	4.826	5.814	6.057	6.162	6.247	6.376	6.578	6.906	7.310	7.860	8.282	8.683	9.492	10.503	11.295
	thereof: $100 \leq x < 500$ kW	90	139	225	370	851	1.928	3.052	4.154	4.475	4.658	4.886	4.981	5.188	5.669	6.402	7.105	7.557	7.964	8.853	9.968	10.845
	thereof: $500 \leq x \leq 750$ kW	6	13	21	34	104	256	426	589	640	662	669	701	848	1.310	1.972	2.609	2.893	3.079	3.379	3.724	3.970
	thereof: $700 < x \leq 1.000$ kW	4	8	11	27	99	200	327	447	468	488	502	595	595	595	592	606	614	648	832	1.131	1.332
	thereof: > 1.000 kW	19	30	46	68	111	182	273	396	454	482	495	611	615	621	617	637	670	907	1.302	1.879	2.489
net electrical capacity	Figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	ground-mounted PV systems	136	195	354	555	1.288	2.792	4.993	8.099	9.056	9.453	10.010	10.521	10.959	11.747	12.482	13.951	15.827	18.460	22.126	27.545	34.695
	thereof: < 10 kW	7	8	9	9	10	13	13	13	13	13	14	14	14	14	14	15	16	17	21	23	24
	thereof: $10 \leq x < 30$ kW	2	2	4	5	6	7	8	8	9	9	9	9	10	10	10	11	11	12	16	20	23
	thereof: $30 \leq x < 100$ kW	18	20	21	23	27	31	35	40	41	43	44	44	45	47	49	52	55	58	64	73	79
	thereof: $100 \leq x \leq 750$ kW	28	41	56	81	139	253	347	505	564	591	616	625	716	1.021	1.338	1.716	1.934	2.039	2.131	2.253	2.372
	thereof: $750 < x \leq 1.000$ kW	6	9	13	27	64	120	181	314	367	389	410	420	425	428	431	454	499	528	637	878	1.236
	thereof: $1.000 < x \leq 10.000$ kW	75	115	251	400	937	2.071	3.778	5.720	6.587	6.956	7.469	7.962	8.304	8.765	9.177	9.974	11.033	12.619	14.134	16.101	18.340
	thereof: > 10.000 kW	-	-	-	10	105	297	631	1.499	1.475	1.452	1.448	1.447	1.445	1.462	1.463	1.729	2.279	3.187	5.123	8.197	12.621
	Total	1.831	2.592	3.741	5.514	9.612	16.508	23.833	31.396	33.801	34.870	36.062	37.360	38.799	41.393	44.657	49.660	54.708	61.515	75.543	91.208	106.096

1) The value includes, in addition to the 1.3 million systems registered in the German Market Master Data Register (MaSiR), an estimate from a research project of a further 2.9 million unregistered plug-in PV systems (commonly referred to as "balcony power plants"). The total installed gross capacity of all balcony power plants amounts to 3.7 GW, of which 1.2 GW are officially registered in the MaSiR. The total net capacity (inverter capacity) of all balcony power plants amounts to 2.5 GW, of which 0.8 GW are registered in the MaSiR.

Table 5: Final energy consumption from renewable sources for heating and cooling

values before 2005 hidden

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Status: February 2026

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Solid biofuels (households) ¹⁾	52.222	61.496	64.092	74.020	65.220	79.304	71.904	83.015	85.941	69.484	69.477	66.289	68.393	70.192	71.154	66.874	76.874	85.440	66.187	71.089	78.088
Solid biofuels (TCS sector) ²⁾	9.695	11.120	11.443	16.046	16.476	21.236	15.813	17.437	18.696	14.945	17.137	16.731	17.450	17.812	19.219	19.101	21.908	19.926	21.429	22.681	24.766
Solid biofuels (industry) ³⁾	21.266	20.319	22.367	20.156	22.972	28.088	29.089	27.793	25.600	26.530	25.108	27.031	26.326	24.522	23.784	23.279	24.820	25.589	25.291	26.324	26.324
Solid biofuels (HP/CHP) ⁴⁾	2.043	2.104	2.225	2.759	3.581	4.057	4.665	5.776	5.532	5.465	5.957	6.259	6.193	5.740	6.121	6.296	6.796	6.470	6.685	6.800	7.021
Liquid biofuels ⁵⁾	1.225	1.814	2.869	3.442	3.735	3.442	2.603	2.204	2.196	2.372	2.189	2.188	2.194	2.298	2.383	3.217	2.591	2.471	2.532	2.184	2.340
Biogas	875	1.495	3.935	3.728	5.412	7.811	9.352	8.903	9.458	10.639	11.588	12.306	12.980	13.304	13.783	14.339	15.015	15.857	14.906	15.944	15.427
Biomethane	0	0	23	68	136	505	760	1.329	2.533	3.268	3.970	4.039	3.740	3.861	3.983	4.216	4.982	4.797	4.982	4.795	4.505
Sewage gas	2.082	1.852	1.858	1.972	1.977	1.999	2.059	2.017	1.805	1.801	2.001	2.050	2.141	2.500	2.402	2.378	2.367	2.322	2.311	2.326	2.326
Landfill gas	231	227	210	154	155	117	101	94	93	98	120	116	107	110	107	95	82	72	65	62	60
Gaseous biofuels	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Biogenic fraction of waste ⁶⁾	7.199	8.433	10.747	6.662	6.530	7.260	8.140	9.033	11.645	11.380	11.807	11.669	12.669	14.508	15.308	15.060	15.601	15.033	14.278	14.623	14.396
Solarthermal energy	2.857	3.363	3.757	4.301	5.061	5.378	6.181	6.416	6.500	7.019	7.562	7.611	7.827	8.963	8.659	8.998	8.565	9.730	9.121	8.804	9.238
Geothermal energy ⁷⁾	532	525	524	550	623	689	722	805	864	1.052	969	1.146	1.168	1.308	1.369	1.427	1.575	1.822	1.796	1.914	2.030
Near-surface geothermal energy, ambient heat ⁸⁾	2.361	2.559	2.811	3.377	3.945	5.013	4.942	5.938	6.871	6.638	7.681	8.564	9.326	9.775	10.801	11.648	13.985	14.581	17.174	19.774	23.327
Total	102.588	115.307	126.861	137.235	135.823	164.899	156.331	170.760	177.734	160.691	165.566	165.999	170.514	174.893	179.073	176.928	195.161	204.110	186.757	197.320	209.848

1) till 2004 according to the Working Group on Energy Balances (AGEB); since 2005 according to Thünen Institute; including charcoal

2) TCS = trade, commerce and service sector; Final energy consumption for heat only production according to Thünen Institute plus fuel input for heat production in decentralised CHP plants; including charcoal; since 2018 including sewage sludge

3) in accordance with Section 8 Energy Statistics Act; including sewage sludge

4) in accordance with Section 3 and 5 Energy Statistics Act, including sewage sludge; HP = heating plants, CHP plant = combined heat and power plant

5) including consumption of biodiesel in agriculture, forestry and military; since 2010 including blended bioethanol

6) biogenic fraction of waste in waste incineration plants estimated at 50 %, since 2008 municipal waste only, decrease 2008 due to first-time inclusion of newly available data (statistical adjustment)

7) before 2003 balneological plants are not taken into account

8) based on GZB and FH-ISE, renewable heat from heat pumps (air-water, water-water, brine-water, process water and gas heat pumps)

Sources: AGEE-Stat based on StBA [26], [28]; ZSW; AGEB [1]; TI [23], [29], [31]; FNR [19]; Uni HH [15]; DENA [24]; DBFZ; LIAG; GZB [2]; FH-ISE [39]; BDH; BSW; DEPV; BWP; partially preliminary data

Table 5.1: Renewable energy sources in the production of district heating

values before 2005 hidden

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Status: February 2026

Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Biogenic fraction of waste ¹⁾	4.378	5.304	5.482	6.373	6.105	6.563	6.728	7.414	8.371	8.225	8.412	8.518	8.993	10.010	10.133	9.628	10.541	10.076	10.295	10.724	10.456
Solid biofuels ²⁾	2.242	2.314	2.438	3.028	3.929	4.428	5.184	6.467	6.210	6.250	6.783	7.169	7.084	6.657	6.895	7.091	7.666	7.353	7.617	7.747	7.990
Gaseous and liquid biofuels ³⁾	272	281	341	343	541	574	728	1.004	869	915	1.127	1.147	1.130	2.348	2.590	2.840	3.099	3.473	3.400	3.864	3.868
Biomethane	-	-	-	-	-	-	-	-	524	951	1.490	1.471	1.404	1.960	2.028	2.235	2.401	2.280	2.127	1.918	1.974
Geothermal energy, ambient heat, solarthermal energy	-	-	-	-	-	-	-	-	61	87	177	223	180	372	484	552	557	913	866	1.130	1.253
Total	6.892	7.899	8.261	9.744	10.575	11.565	12.640	14.885	16.035	16.428	17.989	18.528	18.791	21.347	22.130	22.346	24.264	24.095	24.305	25.383	25.541

Figures in [%]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
for information: share of grid losses and consumption in the transformation sector ⁴⁾	8,9	9,1	8,7	8,9	8,9	8,4	10,0	10,7	10,9	12,6	12,2	12,7	12,6	13,8	11,2	11,2	11,3	12,0	12,2	12,2	12,1

Grid losses are not taken into account

1) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste

2) including sewage sludge

3) sum of gaseous and liquid biomasses: biogas, biomethane, sewage gas, landfill gas and liquid biofuels; including geothermal energy, near-surface geothermal energy, ambient heat and biomethane; reported separately as of 2013

4) share of grid losses including consumption in the transformation sector, with reference to the total district heat production (renewable and fossil energy sources)

Sources: AGEE-Stat based on StBA [26]

Table 5.2: Final energy consumption from renewable sources for heating and cooling by sectors
Energy sources used directly on-site in the sectors for heating and cooling purposes, excluding electricity, excluding district heating consumption

values before 2005 hidden
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Status: February 2026

	Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Total	Solid biofuels ¹⁾	83.183	92.935	97.902	110.221	104.669	128.630	116.806	128.244	130.237	110.957	111.723	110.051	112.169	112.526	114.157	109.255	123.602	130.945	112.906	120.094	129.178
	Gaseous and liquid biofuels ²⁾	4.207	5.181	8.622	9.104	10.987	13.410	14.231	13.658	12.779	14.115	14.912	15.662	16.446	16.182	16.368	17.522	17.300	17.655	16.827	17.122	16.753
	Biomethane	-	-	-	-	-	-	-	-	2.066	2.437	2.661	2.755	2.513	2.172	2.182	2.232	2.853	2.790	3.115	3.112	2.770
	Biogenic fraction of waste ³⁾	3.210	3.611	5.743	855	966	1.245	2.086	2.411	4.188	4.188	4.419	4.233	4.807	5.877	6.314	6.513	6.256	6.168	5.241	5.209	5.209
	Geothermal energy, ambient heat, solarthermal energy	5.715	6.409	7.055	8.163	9.565	11.017	11.831	13.132	14.174	14.622	16.037	17.099	18.140	19.724	20.398	21.585	23.633	25.329	27.332	29.499	33.495
	Total	96.315	108.136	119.322	128.343	126.187	154.302	144.954	157.445	163.444	146.315	149.751	149.800	154.075	156.481	159.419	157.107	173.644	182.891	165.421	175.036	187.405
Households	Solid biofuels ¹⁾	52.222	61.496	64.093	74.020	65.220	79.304	71.904	83.014	85.941	69.483	69.477	66.289	68.393	70.192	71.154	66.874	76.874	85.434	66.185	71.089	78.088
	Gaseous and liquid biofuels ²⁾	162	381	948	943	1.421	1.753	2.055	2.466	2.277	2.591	2.822	3.008	3.176	2.994	3.014	3.109	3.212	3.294	3.072	3.175	3.032
	Biomethane	-	-	-	-	-	-	-	-	554	644	662	697	659	563	574	577	624	691	804	723	611
	Solarthermal energy	2.649	3.138	3.514	4.040	4.772	5.085	5.861	6.094	6.184	6.690	7.218	7.272	7.484	8.573	8.286	8.607	8.189	9.306	8.713	8.409	8.816
	Geothermal energy ⁴⁾	82	50	47	67	95	164	184	245	253	236	267	346	323	267	294	281	441	414	453	487	526
	Near-surface geothermal energy, ambient heat ⁵⁾	2.214	2.400	2.637	3.178	3.700	4.682	4.616	5.552	6.431	6.213	7.174	7.973	8.645	8.909	9.798	10.553	12.667	13.179	15.532	17.754	20.971
	Total	57.329	67.465	71.239	82.248	75.208	90.988	84.620	97.371	101.640	85.857	87.620	85.585	88.680	91.498	93.120	90.001	102.007	112.318	94.760	101.637	112.944
TCS sector	Solid biofuels ¹⁾	9.695	11.120	11.443	16.046	16.476	21.237	15.813	17.437	18.696	14.945	17.137	16.731	17.450	17.812	19.219	19.101	21.908	19.926	21.425	22.681	24.766
	Gaseous biofuels ⁶⁾	2.514	2.846	4.307	4.341	5.572	6.534	7.452	8.503	7.809	8.645	9.436	9.965	10.514	10.351	10.268	10.477	10.753	10.979	10.370	10.667	10.284
	Liquid biofuels, biodiesel in agriculture and forestry	1.110	1.683	2.596	2.591	3.148	2.762	2.137	2.016	1.910	2.066	1.912	1.894	1.935	2.096	2.146	3.030	2.428	2.266	2.349	2.001	2.158
	Biomethane	-	-	-	-	-	-	-	-	1.453	1.725	1.807	1.854	1.736	1.444	1.459	1.407	1.529	1.623	1.889	1.642	1.412
	Solarthermal energy	208	226	243	261	289	293	321	319	312	326	343	338	341	385	369	380	359	403	375	362	380
	Geothermal energy ⁴⁾	416	436	440	418	464	462	523	536	554	732	528	579	666	721	647	666	659	669	660	667	683
	Near-surface geothermal energy, ambient heat ⁵⁾	146	159	174	199	245	331	326	386	440	425	507	591	681	743	875	968	1.177	1.224	1.443	1.649	1.948
	Total ⁷⁾	14.441	16.806	19.516	24.092	26.432	31.861	26.591	29.197	31.174	28.864	31.670	31.952	33.323	33.552	34.983	36.029	38.813	37.090	38.515	39.669	41.631
Industry	Solid biofuels ¹⁾	21.266	20.319	22.366	20.155	22.973	28.089	29.089	27.793	25.600	26.529	25.109	27.031	26.326	24.522	23.784	23.280	24.820	25.589	25.291	26.324	26.324
	Gaseous and liquid biofuels ²⁾	421	271	771	1.229	846	2.361	2.587	673	783	813	742	795	821	741	940	906	907	1.116	1.036	1.279	1.279
	Biomethane	-	-	-	-	-	-	-	-	59	68	192	204	118	165	149	248	700	476	422	747	747
	Biogenic fraction of waste ³⁾	2.858	3.275	5.430	619	728	1.003	2.067	2.411	4.188	4.188	4.419	4.233	4.807	5.877	6.314	6.513	6.256	6.168	5.241	5.209	5.209
	Geothermal energy, ambient heat, solarthermal energy	-	-	-	-	-	-	-	-	-	-	-	-	-	126	129	130	140	134	155	171	171
	Total	24.545	23.865	28.567	22.003	24.547	31.453	33.743	30.877	30.630	31.598	30.462	32.263	32.072	31.431	31.316	31.077	32.823	33.483	32.145	33.730	33.730

1) including sewage sludge

2) sum of gaseous and liquid biomasses: biogas, biomethane, sewage gas, landfill gas and liquid biofuels; since 2013 biomethane depicted separately

3) biogenic fraction of waste in waste incineration plants estimated at 50 %, from 2008 only municipal waste

4) before 2003 balneological plants are not taken into account

5) based on G28 and FH-ISE, renewable heat from heat pumps (air-water, water-water, brine-water, process water and gas heat pumps)

6) sum of biogas, biomethane, sewage gas and landfill gas; biomethane since 2013 depicted separately

7) includes low amounts of energy from the biogenic fraction of waste of the TCS sector

Table 5.3: Additional differentiation: Ground source geothermal energy and environmental heat (heat pumps) ¹

values before 2005 hidden

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Total Number of Units	figures in [1,000 units]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Electric Heating Heat Pumps	117	141	167	203	236	270	310	353	396	438	481	535	596	659	726	812	920	1,061	1,343	1,492	1,725
	of which: Air-to-Water Heat Pumps	17	26	39	57	75	97	126	158	192	225	260	302	350	400	454	525	616	742	1,006	1,145	1,369
	of which: Brine-to-Water Heat Pumps	83	96	108	124	137	148	158	167	175	183	190	200	212	224	236	250	266	280	297	306	315
	of which: Water-to-Water Heat Pumps	17	19	20	22	24	25	26	28	29	30	31	33	34	35	36	37	38	39	40	41	41
	Domestic Hot Water Heat Pumps	229	214	198	190	182	173	166	163	163	166	170	175	183	192	204	226	244	276	354	391	435
	Gas-Driven Heat Pumps	0.1	0.1	0.1	0.2	0.3	0.6	1.0	1.5	2.3	3.3	4.8	6.2	7.2	8.2	9.0	9.8	10.5	11.3	12.0	12.0	12.0
	Total	346	355	365	393	418	444	477	518	561	607	656	716	786	859	939	1,048	1,175	1,348	1,709	1,895	2,172

Total Thermal Capacity	figures in [MW]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Electric Heating Heat Pumps	1,629	1,897	2,177	2,549	2,904	3,337	3,813	4,315	4,781	5,204	5,642	6,160	6,748	7,362	7,994	8,821	9,855	11,209	13,944	15,371	17,605
	of which: Air-to-Water Heat Pumps	247	360	515	727	938	1,235	1,577	1,959	2,321	2,646	2,995	3,398	3,853	4,329	4,835	5,513	6,383	7,586	10,129	11,462	13,605
	of which: Brine-to-Water Heat Pumps	1,090	1,224	1,330	1,465	1,584	1,700	1,815	1,914	1,999	2,081	2,150	2,245	2,357	2,475	2,589	2,725	2,876	3,013	3,186	3,268	3,349
	of which: Water-to-Water Heat Pumps	292	313	332	357	382	402	421	442	461	477	497	517	538	558	570	583	596	610	629	641	651
	Domestic Hot Water Heat Pumps	458	427	395	380	363	345	332	325	325	332	340	350	365	385	409	451	489	552	708	781	870
	Gas-Driven Heat Pumps	4	6	7	9	16	27	42	63	89	123	172	216	246	276	301	324	347	369	390	388	385
	Total	2,091	2,330	2,579	2,938	3,283	3,709	4,187	4,703	5,195	5,659	6,154	6,726	7,359	8,023	8,704	9,596	10,691	12,130	15,042	16,540	18,860

Thermal Work / Total Heat	figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Electric Heating Heat Pumps	3,282	3,553	3,883	4,743	5,583	7,320	6,923	8,485	9,961	9,144	10,750	12,042	13,042	13,415	14,854	15,884	19,613	19,761	23,075	26,418	31,590
	of which: Air-to-Water Heat Pumps	471	616	843	1,262	1,725	2,576	2,747	3,717	4,711	4,564	5,601	6,527	7,324	7,771	8,853	9,761	12,475	13,089	16,239	19,449	24,009
	of which: Brine-to-Water Heat Pumps	2,214	2,328	2,427	2,794	3,107	3,832	3,385	3,874	4,268	3,726	4,187	4,482	4,652	4,601	4,909	5,032	5,898	5,539	5,698	5,825	6,345
	of which: Water-to-Water Heat Pumps	597	609	613	687	751	912	791	894	982	854	962	1,033	1,066	1,043	1,092	1,091	1,240	1,133	1,138	1,144	1,236
	Domestic Hot Water Heat Pumps	272	251	233	219	211	202	194	189	188	191	196	203	211	222	235	255	279	310	375	444	492
	Gas-Driven Heat Pumps	6	10	13	17	26	43	70	107	156	218	303	398	474	535	591	641	689	735	780	803	806
	Total	3,560	3,814	4,129	4,979	5,820	7,565	7,187	8,781	10,305	9,553	11,249	12,643	13,727	14,172	15,680	16,780	20,581	20,806	24,230	27,665	32,888

Final Energy Consumption Electricity / Gas	figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Electric Heating Heat Pumps	1,081	1,144	1,214	1,504	1,775	2,445	2,123	2,697	3,252	2,690	3,281	3,723	3,989	3,940	4,381	4,593	6,018	5,602	6,384	7,162	8,814
	of which: Air-to-Water Heat Pumps	207	251	316	479	651	1,030	953	1,352	1,767	1,458	1,880	2,220	2,443	2,443	2,790	2,986	4,107	3,871	4,628	5,388	6,866
	of which: Brine-to-Water Heat Pumps	688	707	716	821	904	1,141	947	1,091	1,206	1,001	1,138	1,221	1,257	1,220	1,300	1,319	1,577	1,435	1,461	1,480	1,628
	of which: Water-to-Water Heat Pumps	186	186	182	204	220	274	223	254	279	231	263	282	289	277	291	288	334	296	295	294	320
	Domestic Hot Water Heat Pumps	114	104	95	87	82	77	72	69	67	66	67	68	70	73	76	81	88	96	114	133	146
	Gas-Driven Heat Pumps	4	6	9	11	17	31	50	78	113	159	220	288	342	384	423	457	490	521	551	567	568
	Total	1,199	1,254	1,318	1,602	1,874	2,553	2,245	2,844	3,432	2,915	3,568	4,079	4,401	4,397	4,880	5,131	6,596	6,219	7,049	7,862	9,528

Renewable Heat	figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Electric Heating Heat Pumps	2,201	2,409	2,669	3,240	3,808	4,875	4,801	5,789	6,708	6,454	7,469	8,320	9,053	9,474	10,473	11,291	13,594	14,159	16,691	19,254	22,777
	of which: Air-to-Water Heat Pumps	265	365	527	783	1,074	1,546	1,794	2,365	2,944	3,106	3,721	4,307	4,881	5,328	6,063	6,775	8,368	9,218	11,611	14,060	17,143
	of which: Brine-to-Water Heat Pumps	1,525	1,621	1,711	1,973	2,203	2,691	2,439	2,783	3,061	2,725	3,049	3,262	3,395	3,381	3,608	3,713	4,320	4,104	4,237	4,344	4,718
	of which: Water-to-Water Heat Pumps	411	423	431	484	531	638	568	641	703	623	699	751	777	765	802	803	906	837	843	850	916
	Domestic Hot Water Heat Pumps	157	147	138	132	129	125	122	120	121	125	129	135	141	149	159	174	191	214	261	311	346
	Gas-Driven Heat Pumps	2	3	4	6	8	13	20	29	42	59	83	110	132	151	168	184	199	214	229	236	237
	Total	2,360	2,559	2,811	3,378	3,945	5,013	4,943	5,938	6,871	6,638	7,681	8,565	9,326	9,774	10,800	11,649	13,984	14,587	17,181	19,801	23,360

¹⁾ based on G2B and FH-ISE

Table 6: Final energy consumption from renewable sources in the transport sector ¹⁾

values before 2005 hidden

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Figures in [GWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Biodiesel including HVO	17.666	27.938	32.282	25.873	22.966	24.359	23.556	24.628	21.945	22.676	20.829	20.896	21.354	22.329	22.109	30.148	25.097	24.817	25.626	21.583	24.013
thereof: Biodiesel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	21.610	21.005	21.507	22.145	18.725	20.861
thereof: blended Biodiesel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	21.559	20.933	21.456	22.083	18.663	20.799
thereof: neat Biodiesel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	51	72	51	62	62	62
thereof: HVO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8.538	4.092	3.310	3.481	2.858	3.152
thereof: blended HVO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8.514	4.019	3.200	3.066	1.710	1.527
thereof: neat HVO	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	24	73	110	415	1.148	1.625
Vegetable oil	1.828	7.206	8.533	4.042	961	574	188	251	0	52	10	31	31	10	21	21	10	10	10	0	0
Bioethanol	1.780	3.828	3.391	4.608	6.576	8.537	9.031	9.149	8.832	9.002	8.589	8.604	8.464	8.685	8.353	8.014	8.412	8.692	9.142	9.142	9.496
Biomethane ²⁾	0	0	0	4	13	75	92	333	483	449	345	379	445	389	660	884	965	1.332	1.871	3.442	3.898
thereof: CNG	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	884	965	1.304	1.338	1.400	1.500
thereof: LNG	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0	28	533	2.042	2.398
RFNBO ³⁾	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5
thereof: renewable hydrogen	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5
RE electricity consumption in transport ⁴⁾	1.353	1.484	1.762	1.699	1.925	2.077	2.494	2.862	3.017	3.169	3.523	3.732	4.329	4.581	4.898	5.237	5.366	6.524	8.102	9.222	10.329
thereof: road transport	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	578	1.137	1.972	3.214	4.010
thereof: rail transport	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4.788	5.387	6.130	6.008	6.319
Total	22.627	40.456	45.968	36.226	32.441	35.622	35.361	37.223	34.277	35.348	33.296	33.642	34.623	35.994	36.041	44.304	39.850	41.375	44.751	43.389	47.741

1) consumption of fuels in the transport sector (excluding consumption in agriculture, forestry and military)
 2) until 2021 exclusively Bio CNG, from 2022 Bio CNG and Bio LNG; based on the net calorific value
 3) Renewable Fuels of Non-Biological Origin (RFNBO), in accordance with EU Directive 2018/2001, Delegated Regulation (EU) 2023/1184, and 37th BImSchV
 4) calculated on the share of renewables-based electricity generation in gross electricity consumption in each year

Sources: AGEE-Stat based on BAFA [5]; BLE [3]; [4]; BMF [6]; The German Gas and Hydrogen Industry [34]; StBA [25]; UBA [33]; FNR; ZSW; DBFZ; AGQM; UFOP; partially preliminary data

Table 7: Development of energy consumption in Germany

values before 2005 hidden

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Figures in [TWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Gross electricity consumption ¹⁾	618,3	622,9	624,9	621,5	584,5	618,9	610,4	610,1	607,2	594,5	600,7	600,6	601,9	594,8	577,6	557,9	574,3	551,7	521,0	529,4	526,9
Production of district heating	136,1	135,8	130,1	133,0	130,4	143,1	129,8	133,6	135,6	121,8	127,1	130,3	130,5	130,2	127,2	118,3	137,8	119,7	115,4	119,7	122,8
Final energy consumption for heating and cooling (including district heat) ²⁾	1.275,3	1.341,3	1.199,7	1.313,5	1.208,6	1.347,0	1.239,6	1.259,1	1.306,8	1.175,1	1.217,7	1.238,3	1.241,3	1.210,1	1.229,0	1.201,0	1.266,9	1.167,6	1.075,7	1.086,8	1.106,2
Final energy consumption for heating and cooling (without electricity and district heat) ³⁾	1.151,5	1.217,7	1.081,0	1.192,3	1.089,8	1.215,9	1.122,9	1.139,9	1.186,2	1.068,7	1.106,1	1.124,6	1.127,2	1.097,6	1.115,5	1.095,1	1.143,9	1.062,3	974,4	980,0	996,3
thereof in: Households	538,3	553,8	453,5	542,3	505,8	550,8	468,1	508,8	538,9	448,5	472,0	485,4	480,4	484,7	512,6	505,3	512,2	486,1	441,5	445,9	463,0
thereof in: TCS	191,6	230,0	180,7	207,0	195,5	209,4	191,3	188,3	211,8	192,6	199,5	190,5	192,1	169,1	172,5	172,8	177,3	164,6	151,5	151,1	160,3
thereof in: Industry	421,6	433,9	446,8	443,0	388,5	455,7	463,5	442,8	435,5	427,6	434,6	448,7	454,7	443,8	430,4	417,0	454,4	411,6	381,4	383,0	373,0
Final energy consumption in the transport sector ⁴⁾	634,0	654,9	622,5	610,7	606,2	611,9	617,5	612,4	623,9	630,0	639,8	650,6	662,8	646,6	649,9	591,2	591,9	598,8	588,5	587,5	599,4

Figures in [PJ]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Primary energy consumption ⁵⁾	14.506	14.852	14.198	14.381	13.513	14.137	13.523	13.630	13.916	13.244	13.384	13.510	13.528	13.190	12.821	11.891	12.465	11.692	10.633	10.571	10.490

1) gross electricity production by fossil fuels accounting to AGEb and AGEE-Stat, status February 2026, data on electricity trade according to StBA

2) sectoral local consumption of energy for application purposes of heating and cooling, without electricity consumption, including district heat; calculation based on Working Group on Energy Balances (AGEb) and AGEE-Stat, status February 2026

3) sectoral local consumption of energy for application purposes of heating and cooling, without secondary energy carriers electricity and district heat; calculation based on Working Group on Energy Balances (AGEb) and AGEE-Stat, status February 2026

4) calculation based on Working Group on Energy Balances (AGEb) and AGEE-Stat, without energy consumption for international aviation, status February 2026

5) calculation based on Working Group on Energy Balances (AGEb) and AGEE-Stat, calculated by the "physical energy content" principle, status Februar 2026

Sources: AGEE-Stat based on AGEb [1]; StBA [30]; Eurostat [18]; IEA [22]; EP/ER [16], [17]; partially preliminary data

Table 8.1: Greenhouse gas emissions avoided through the use of renewable energy sources ¹

values before 2005 hidden

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Total	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	by RE gross electricity production	56.052	55.705	66.096	65.196	67.903	74.158	91.539	92.835	95.743	112.945	129.309	129.220	142.160	151.278	177.451	184.228	172.569	184.328	198.960	206.684	207.259
	by RE final energy consumption for heating and cooling	20.351	22.314	24.731	26.836	27.317	32.158	31.002	33.291	34.149	31.575	32.845	33.249	33.304	36.109	36.568	36.248	40.086	42.730	38.997	40.328	42.666
	by RE final energy consumption in the transport sector ^{1) 2)}	4.248	7.684	8.728	6.913	6.117	6.663	6.540	7.077	6.478	6.746	6.359	6.972	7.270	7.594	7.380	11.198	10.460	11.257	12.925	13.711	15.475
	Total avoided GHG-emissions	80.651	85.703	99.555	98.945	101.337	112.979	129.081	133.203	136.370	151.266	168.513	169.441	182.734	194.981	221.399	231.674	223.115	238.315	250.882	260.723	265.400

by RE gross electricity production	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Hydropower	20.794	17.367	17.850	16.140	15.417	16.898	14.855	16.766	16.402	15.584	14.887	15.878	15.022	13.815	16.402	15.192	15.929	14.328	16.072	18.105	13.625
	Wind energy onshore	22.487	23.445	30.137	28.812	28.482	27.654	38.006	33.854	36.733	43.563	53.486	49.831	61.683	65.372	77.518	79.970	70.052	75.648	89.357	85.861	80.938
	Wind energy offshore	0	0	0	0	28	127	429	493	653	1.138	6.068	9.121	12.544	14.207	19.279	21.236	18.857	19.497	18.425	20.022	20.325
	Solar Photovoltaic	602	1.345	1.857	3.050	4.624	8.222	14.320	16.766	18.288	23.618	25.555	25.047	24.958	28.494	31.969	34.997	35.080	42.694	44.396	51.958	62.718
	Solid biofuels	6.328	6.827	6.618	6.472	6.987	7.568	7.984	7.499	6.894	7.867	7.869	7.632	7.225	7.746	8.370	8.558	8.070	8.036	7.878	7.711	7.613
	Liquid biofuels	82	414	544	613	959	743	229	187	140	189	234	270	226	203	183	147	76	24	25	14	11
	Biogas	772	1.572	3.950	5.129	6.563	7.665	10.108	11.715	10.943	13.676	14.007	14.116	13.549	14.014	15.887	16.233	16.525	16.427	15.079	15.598	14.826
	Biomethane	0	0	10	22	41	192	316	523	717	1.245	1.474	1.476	1.325	1.425	1.641	1.693	1.813	1.703	1.732	1.556	1.513
	Sewage gas	1.006	695	676	710	760	802	883	836	760	880	897	926	903	1.001	1.122	1.128	1.118	1.098	1.088	1.090	1.092
	Landfill gas	981	719	661	560	529	449	433	341	280	287	256	230	209	196	202	173	173	142	146	135	108
	Biogenic fraction of waste	3.000	3.321	3.793	3.678	3.502	3.821	3.964	3.841	3.892	4.840	4.498	4.591	4.424	4.701	4.748	4.746	4.715	4.597	4.632	4.489	4.338
	Geothermal energy	0	0	0	10	11	17	12	14	41	58	78	102	92	104	130	155	161	134	130	145	152
	Total	56.052	55.705	66.096	65.196	67.903	74.158	91.539	92.835	95.743	112.945	129.309	129.220	142.160	151.278	177.451	184.228	172.569	184.328	198.960	206.684	207.259

by RE final energy consumption for heating and cooling	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Solid biofuels (households)	7.460	8.831	9.379	11.031	9.836	11.864	10.835	12.673	13.302	11.005	10.733	10.377	10.675	11.527	11.397	10.698	12.613	14.207	10.912	11.501	12.641
	Solid biofuels (TCS sector)	2.451	2.847	2.941	4.084	4.177	5.318	4.039	4.479	4.847	3.826	4.271	4.185	3.720	4.018	4.214	4.120	4.827	4.542	4.707	4.825	5.229
	Solid biofuels (industry)	6.228	5.860	6.461	6.171	6.991	7.938	8.404	7.860	6.953	7.251	7.213	7.743	7.387	7.408	6.954	6.691	7.129	7.706	7.178	7.226	7.205
	Solid biofuels (HP/CHP)	498	484	544	675	887	981	1.138	1.420	1.203	1.205	1.397	1.528	1.480	1.414	1.518	1.577	1.693	1.693	1.752	1.783	1.836
	Liquid biofuels	269	406	641	716	800	720	533	456	446	487	480	509	549	577	576	716	569	541	506	423	460
	Biogas	80	136	362	358	513	781	990	868	928	1.050	1.568	1.692	1.878	2.323	2.304	2.321	2.566	2.794	2.526	2.661	2.569
	Biomethane	0	0	3	10	20	76	118	187	365	471	883	815	782	548	618	646	739	735	715	645	609
	Sewage gas	505	446	444	475	468	478	502	464	412	411	510	522	553	664	673	697	732	728	716	712	713
	Landfill gas	70	69	64	47	47	36	32	28	27	29	41	41	38	41	40	35	29	25	20	19	19
	Gaseous biofuels	1.649	1.926	2.454	1.528	1.513	1.644	1.857	2.074	2.665	2.611	2.471	2.352	2.475	3.061	3.451	3.556	3.707	3.730	3.504	3.589	3.533
	Solarthermal energy	791	927	1.030	1.186	1.403	1.533	1.763	1.829	1.854	2.002	1.981	1.988	2.043	2.483	2.388	2.473	2.357	2.722	2.482	2.344	2.460
	Geothermal energy	157	155	154	164	188	202	213	240	256	314	278	322	327	380	400	410	449	542	534	573	608
	Near-surface geothermal energy, ambient heat	193	227	254	391	474	587	578	713	891	913	1.019	1.175	1.397	1.665	2.035	2.308	2.676	2.765	3.445	4.027	4.784
	Total	20.351	22.314	24.731	26.836	27.317	32.158	31.002	33.291	34.149	31.575	32.845	33.249	33.304	36.109	36.568	36.248	40.086	42.730	38.997	40.328	42.666

by RE final energy consumption in the transport sector ^{3) 4)}	Figures in [1.000 t CO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Biodiesel incl. HVO	3.480	5.462	6.357	5.123	4.490	4.669	4.594	5.065	4.516	4.731	4.432	4.904	5.118	5.347	5.118	8.200	7.092	7.181	7.488	6.298	7.007
	Vegetable oil	374	1.366	1.598	761	189	115	34	45	0	9	2	6	6	2	4	5	2	2	2	0	0
	Bioethanol	394	856	773	1.028	1.435	1.846	1.895	1.902	1.867	1.915	1.855	1.978	2.046	2.160	2.132	2.429	2.505	2.580	2.740	2.710	2.815
	Biomethane	0	0	0	1	3	33	17	65	95	91	70	84	100	85	126	270	306	574	1.073	1.963	2.223
	RE electricity consumption in transport ⁴⁾	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	294	555	920	1.622	2.740	3.430
	Total	4.248	7.684	8.728	6.913	6.117	6.663	6.540	7.077	6.478	6.746	6.359	6.972	7.270	7.594	7.380	11.198	10.460	11.257	12.925	13.711	15.475

1) not including biogenic CO2 from the thermal use of wood based solid biofuels

2) consumption of biogenic fuels in the transport sector (excluding consumption in agriculture, forestry and military and excluding electricity consumption in the transport sector)

3) biofuels based on data of the Federal Office for Agriculture and Food (BLE) for the year 2023 and the fossil reference values according § 3 and § 10 of 38. BImSchV

4) accounting of emission reductions from the use of electricity in road transport, taking into account the current electricity mix

Source: UBA [30] based on sources quoted there; partially preliminary data

Table 8.2: Acidifying emissions avoided through the use of renewable energy sources ²

values before 2005 hidden

Status: February 2026

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Figures in [1.000 t SO ₂ -eq.]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
by RE gross electricity production	33,6	25,9	23,9	23,9	21,6	23,3	32,5	21,5	19,5	30,0	97,2	82,2	99,1	106,7	124,0	122,2	114,6	129,8	150,6	156,6	158,8
by RE final energy consumption for heating and cooling	-1,6	-2,8	-4,2	0,9	-2,0	-5,2	-5,5	-8,0	-11,8	-9,8	-13,2	-14,0	-21,2	-21,4	-22,5	-26,2	-27,6	-20,8	-21,4	-23,8	-24,5
by RE final energy consumption in the transport sector ¹⁾	-1,2	-3,6	-11,9	-7,8	-4,8	-7,0	-4,6	-6,0	-6,9	-4,8	-3,7	-3,6	-19,0	-18,6	-17,4	-7,0	-5,6	-3,9	-1,7	-0,3	-0,3
Total avoided acidifying agents ²⁾	30,8	19,5	7,8	17,0	14,8	11,1	22,4	7,5	0,8	15,4	80,3	64,6	58,9	66,7	84,1	89,0	81,4	105,1	127,5	132,5	134,0

1) consumption of biogenic fuels in the transport sector (excluding consumption in agriculture, forestry and military and excluding electricity consumption in the transport sector)

2) as acidifying emissions only SO₂ and Nox are taking into account

Source: UBA [30] based on sources quoted there; partially preliminary data

Table 8.3: Avoidance factors of renewable energy sources ²

values before 2005 hidden

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Status: February 2026

Figures in [g CO ₂ -eq./kWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Avoidance factor of RE gross electricity production	882	765	734	686	700	697	729	640	623	690	680	676	653	671	728	727	720	721	722	717	714
Avoidance factor of RE final energy consumption for heating and cooling	202	196	198	198	204	198	201	198	195	199	201	203	198	209	207	207	207	210	210	205	204
Avoidance factor of RE final energy consumption in the transport sector ^{1) 2)}	200	197	197	200	200	199	199	206	207	210	214	233	240	242	237	279	287	297	308	321	322

Figures in [g SO ₂ -eq./kWh]	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Avoidance factor of RE gross electricity production	0,53	0,36	0,27	0,25	0,22	0,22	0,26	0,15	0,13	0,18	0,51	0,43	0,46	0,47	0,51	0,48	0,48	0,51	0,55	0,54	0,55
Avoidance factor of RE final energy consumption for heating and cooling	-0,02	-0,03	-0,03	0,01	-0,01	-0,03	-0,04	-0,05	-0,07	-0,06	-0,08	-0,09	-0,13	-0,12	-0,13	-0,15	-0,14	-0,10	-0,12	-0,12	-0,12
Avoidance factor of RE final energy consumption in the transport sector ¹⁾	-0,06	-0,09	-0,27	-0,22	-0,16	-0,21	-0,14	-0,17	-0,22	-0,15	-0,12	-0,12	-0,63	-0,59	-0,56	-0,18	-0,16	-0,11	-0,05	-0,01	-0,01

1) consumption of biogenic fuels in the transport sector (excluding consumption in agriculture, forestry and military and excluding electricity consumption in the transport sector)

2) based on data of the Federal Office for Agriculture and Food (BLE) for the year 2023 and the fossil reference values according § 3 and § 10 of 38. BImSchV

3) as acidifying emissions only SO₂ and Nox are taking into account

Source: UBA [30] based on sources quoted there; partially preliminary data

Table 9.1: Investments in construction of renewable energy plants

values before 2005 hidden

Status: February 2026

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Figures in [million euro] ¹⁾	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Hydropower	240	220	330	370	500	350	300	200	130	90	80	60	60	120	110	100	70	70	10	6	4
Wind energy onshore	2.490	3.220	2.470	2.540	2.800	2.110	2.860	3.550	4.490	7.060	5.370	6.910	7.450	3.390	1.650	2.190	2.990	3.830	5.960	5.470	8.820
Wind energy offshore	-	-	30	170	470	450	610	2.440	4.270	3.940	3.680	3.370	3.400	4.100	2.130	80	280	1.510	1.880	3.580	2.530
Solar Photovoltaic	4.840	4.010	5.330	7.970	13.570	19.580	15.860	11.980	3.380	1.450	1.480	1.570	1.660	2.580	3.370	4.920	5.230	8.140	18.540	17.020	15.010
Solar thermal energy	630	990	760	1.700	1.490	990	1.060	950	860	790	800	700	540	490	440	530	550	690	420	240	170
Geothermal energy, ambient heat	410	940	920	1.230	1.140	960	990	1.060	1.090	1.080	1.010	1.210	1.320	1.520	1.410	1.920	2.530	4.590	8.850	5.360	8.640
Biomass electricity	1.910	2.270	2.280	1.980	2.020	2.240	3.120	790	700	670	220	270	280	390	350	320	250	250	290	330	300
Biomass heat	1.510	2.300	1.500	1.760	1.610	1.210	1.320	1.500	1.560	1.320	1.290	1.230	1.230	1.240	1.260	1.940	2.730	3.650	2.770	1.770	2.080
Total	12.030	13.950	13.620	17.720	23.600	27.890	26.120	22.470	16.480	16.400	13.930	15.320	15.940	13.830	10.720	12.000	14.630	22.730	38.720	33.776	37.554

1) differences in the totals may occur due to roundings

Source: ZSW [32]; partially preliminary data

Table 9.2: Economic stimuli from the operation of renewable energy plants

values before 2005 hidden

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Status: February 2026

Figures in [million euro] ¹⁾	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Hydropower	130	130	140	150	160	170	190	190	200	200	200	210	210	210	220	230	230	230	240	240	240
Wind energy onshore	550	630	710	790	870	970	1.060	1.200	1.360	1.550	1.730	1.890	2.080	2.230	2.300	2.310	2.310	2.300	2.280	2.320	2.480
Wind energy offshore	-	-	-	-	10	20	30	60	130	210	280	350	420	500	560	600	620	650	760	940	1.120
Solar Photovoltaic	130	190	260	360	530	770	1.040	1.250	1.360	1.400	1.420	1.440	1.470	1.500	1.540	1.600	1.670	1.760	1.950	2.210	2.390
Solar thermal energy	50	70	90	110	140	170	190	210	230	240	260	270	290	300	310	320	330	340	350	360	360
Geothermal energy, ambient heat	240	290	360	440	530	620	730	820	900	870	1.010	1.120	1.220	1.280	1.410	1.520	1.820	1.900	2.240	2.570	3.020
Biomass electricity	720	1.100	1.680	2.020	2.420	2.880	3.320	4.080	4.200	4.500	4.650	4.640	4.670	4.670	4.780	4.830	4.650	4.850	4.890	4.920	4.720
Biomass heat	1.460	1.740	1.960	2.150	2.450	2.880	2.870	3.120	3.320	3.030	3.180	3.360	3.390	3.340	3.350	3.370	3.830	5.120	5.250	4.780	3.840
Biomass fuels	1.790	3.150	3.750	3.530	2.390	2.920	3.690	3.720	3.050	2.640	2.440	2.560	2.710	2.700	2.830	3.540	4.970	6.750	5.150	4.290	4.720
Total	5.070	7.300	8.950	9.550	9.500	11.400	13.120	14.650	14.750	14.640	15.170	15.840	16.460	16.730	17.300	18.320	20.430	23.900	23.110	22.630	22.890

1) differences in the totals may occur due to roundings

Source: ZSW [32]; partially preliminary data

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		RE 2024	Share of renewable energy	avoided GHG-emissions
		[GWh]	[%]	[1.000 t CO ₂ -eq.]
Gross electricity production	Hydropower	22.481	4,2	18.105
	Wind energy onshore	113.749	21,5	85.861
	Wind energy offshore	26.064	4,9	20.022
	Solar Photovoltaic	75.878	14,3	51.958
	Solid biofuels & sewage sludge	10.259	1,9	7.711
	Liquid biofuels	101	< 0,1	14
	Biogas	29.268	5,5	15.598
	Biomethane	2.796	0,5	1.556
	Sewage gas	1.531	0,3	1.090
	Landfill gas	198	< 0,1	135
	Biogenic fraction of waste	5.544	1,0	4.489
	Geothermal energy	215	< 0,1	145
	Total	288.084	54,4	206.684
Final energy consumption for heating and cooling	Solid biofuels & charcoal (households)	71.089	6,5	11.501
	Solid biofuels & charcoal (TCS sector)	22.681	2,1	4.825
	Solid biofuels & sewage sludge (industry)	26.324	2,4	7.226
	Solid biofuels & sewage sludge (HP/CHP)	6.800	0,6	1.783
	Liquid biofuels	2.184	0,2	423
	Biogas	15.944	1,5	2.661
	Biomethane	4.795	0,4	645
	Sewage gas	2.326	0,2	712
	Landfill gas	62	< 0,1	19
	Biogenic fraction of waste	14.623	1,3	3.589
	Solar thermal energy	8.804	0,8	2.344
	Deep geothermal energy	1.914	0,2	573
	Near-surface geothermal energy & ambient heat	19.774	1,8	4.027
	Total	197.320	18,2	40.328
Final energy consumption transport	Biodiesel incl. HVO	21.583	3,7	6.298
	Vegetable oil	0	0	0
	Bioethanol	9.142	1,6	2.710
	Biomethane	3.442	0,6	1.963
	RE electricity consumption transport	9.222	1,6	2.740
	Total	43.389	7,4	13.711

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		RE 2025	Share of renewable energy	avoided GHG-emissions
		[GWh]	[%]	[1.000 t CO ₂ -eq.]
Gross electricity production	Hydropower	16.897	3,2	13.625
	Wind energy onshore	107.225	20,4	80.938
	Wind energy offshore	26.458	5,0	20.325
	Solar Photovoltaic	91.591	17,4	62.718
	Solid biofuels & sewage sludge	10.126	1,9	7.613
	Liquid biofuels	82	< 0,1	11
	Biogas	27.785	5,3	14.826
	Biomethane	2.713	0,5	1.513
	Sewage gas	1.533	0,3	1.092
	Landfill gas	158	< 0,1	108
	Biogenic fraction of waste	5.357	1,0	4.338
	Geothermal energy	225	< 0,1	152
	Total	290.150	55,1	207.259
Final energy consumption for heating and cooling	Solid biofuels & charcoal (households)	78.088	7,1	12.641
	Solid biofuels & charcoal (TCS sector)	24.766	2,2	5.229
	Solid biofuels & sewage sludge (industry)	26.324	2,4	7.205
	Solid biofuels & sewage sludge (HP/CHP)	7.021	0,6	1.836
	Liquid biofuels	2.340	0,2	460
	Biogas	15.427	1,4	2.569
	Biomethane	4.505	0,4	609
	Sewage gas	2.326	0,2	713
	Landfill gas	60	< 0,1	19
	Biogenic fraction of waste	14.396	1,3	3.533
	Solar thermal energy	9.238	0,8	2.460
	Deep geothermal energy	2.030	0,2	608
	Near-surface geothermal energy & ambient heat	23.327	2,1	4.784
	Total	209.848	19,0	42.666
Final energy consumption transport	Biodiesel incl. HVO	24.013	4,0	7.007
	Vegetable oil	0	0	0
	Bioethanol	9.496	1,6	2.815
	Biomethane	3.898	0,7	2.223
	RFNBO	5	-	-
	RE electricity consumption transport	10.329	1,7	3.430
	Total	47.741	8,0	15.475

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Methodological changes

National and international reporting obligations increase the demand of reliable and long-term updateable statistical data on the development of all renewable energy sources in Germany. The Working Group on Renewable Energy Statistics (AGEE-Stat) continuously improves and unifies data base and calculation methods. If new information allow conclusions to be drawn on historic values, these data are complemented or corrected. An overview of the implemented methodological changes in comparison to the previous publication is provided below:

In 2025, renewable fuels of non-biological origin (RFNBO) certified in accordance with the Renewable Energy Directive (the specific requirements are laid down in the 37th Federal Immission Control Ordinance – 37th BImSchV) were used in the transport sector for the first time. They are now included in „Table 6: Final energy consumption from renewable sources in the transport sector“. At present, the reported volumes consist exclusively of renewable hydrogen.

Conversion factors

Joule	J	for energy, work, heat
Watt	W	for capacity, energy flow, heat flow
1 Joule (J) = 1 Newton metre (Nm) = 1 Watt second (Ws)		

Legally binding units in Germany since 1978.

Calorie and derived units such as coal equivalent and oil equivalent are still used as alternatives.

<i>referred to net calorific value</i>		PJ	TWh	Mtce	Mtoe
1 Petajoule	PJ	1	0,2778	0,0341	0,0239
1 Terawatt hour	TWh	3,6	1	0,123	0,0861
1 million tonne of hard coal unit	Mtce	29,308	8,14	1	0,7
1 million tonne of oil equivalent	Mtoe	41,869	11,63	1,429	1

1 TWh = 1 billion kWh	Kilo (k)	10 ³	Tera (T)	10 ¹²
1 GWh = 1 million kWh	Mega (M)	10 ⁶	Peta (P)	10 ¹⁵
1 MWh = 1.000 kWh	Giga (G)	10 ⁹	Exa (E)	10 ¹⁸

Abbreviations

eq.	Equivalent
RE	Renewable energies
EEG	Renewable Energy Sources Act (Erneuer-bare-Energien-Gesetz)
FEC	Final energy consumption
N/A	Not quantified
PEC	Primary energy consumption
HP	Heating plant
CHP	Combined heat and power plant

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INSTITUTIONS

AGEB	Working Group on Energy Balances e.V., Berlin.
BAFA	Federal Office for Economic Affairs and Export Control, Eschborn.
BDEW	German Association of Energy and Water Industries e.V., Berlin.
BLE	Federal Office of Agriculture and Food, Bonn.
BMEL	Federal Ministry of Food and Agriculture, Bonn.
BMU	Federal Ministry for the Environment, Nature Conservation and Nuclear Safety, Berlin.
BMWi	Federal Ministry for Economic Affairs and Energy, Berlin.
BNetzA	Federal Network Agency, Bonn.
BSW	German Solar Association, Berlin.
BWE	German Wind Energy Association e.V., Berlin.
BWP	German Heat Pump Association e.V., Berlin.
DBFZ	German Biomass Research Centre, Leipzig.
DENA	German Energy Agency, Berlin.
DEPV	German Energy Pellet Association e.V., Berlin.
DIW	German Institute for Economic Research, Berlin.
FNR	Specialist agency renewable raw materials e.V., Gülzow.
GtV	Geothermal Energy Association e.V., Berlin.
GZB	International Geothermal Center, Bochum
LIAG	Leibniz Institute for Applied Geophysics: Geothermal Information System for Germany, www.geotis.de .
RWI	Leibniz Institute for Economic Research, Essen.
StBA	Federal Statistical Office, Wiesbaden.
TI	Institute of International Forestry and Forest Economics, Hamburg.
UBA	Federal Environment Agency, Dessau-Roßlau.
UL	UL International GmbH, DEWI, Wilhelmshaven.
ÜNB	Information platform of the German Transmission System Operators, www.netztransparenz.de
ZSW	Centre for Solar Energy and Hydrogen Research Baden-Württemberg, Stuttgart.

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